

EXP-04: THE MICROPHONE VALLEY

EITT on Canonical Transfer Functions

Bessel Bandpass Filters as Microphone Models
The Slope of EITT as a Compositional Phase-Locked Loop
Stored Energy Attack: EITT Detects Its Own Corruption

Entropy-Invariant Time Transformer (EITT)
Higgins Iterative Validation Protocol (HIVP)
F17 Linear Contamination Tuner

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HIVP Chain: EXP-01 (Gold Test) → EXP-02 (US Monthly) → EXP-03 (Uranium Test) → **EXP-04 (Microphone Valley)**

Complete record including: Bessel analysis, slope/PLL framework, stored energy attack,
Rayleigh pattern discovery, lock range analysis, error correction investigation,
and cross-domain comparison across all four HIVP experiments.

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1. EXECUTIVE SUMMARY

EXP-04 returns EITT to its native domain: acoustics and signal processing. A microphone's frequency response is modeled as a canonical Bessel bandpass filter (HP at 40 Hz, LP at 16 kHz), and the power response at each frequency is decomposed into a signal/loss 2-simplex — a CoDa-legitimate composition that walks from loss-dominated at DC, through the transduction passband, to loss-dominated at extreme HF.

The experiment tests Bessel orders 1 through 6, compares Butterworth, Chebyshev I/II, and elliptic approximations, and isolates the passband, LF rolloff, and HF rolloff regions independently. 500 log-spaced frequency points from 1 Hz to 48 kHz provide the walk.

What began as a simple decimation test produced three unexpected discoveries:

Discovery	Section	Significance
EITT Slope as PLL	4-5	The slope $d(\text{rel}\%)/dM$ is a constraint detector — flat = in lock, rising = losing lock
Stored Energy Attack	7	EITT detects its own contamination. F17 C_geom alarm triggers on entropy stuffing
Rayleigh Pattern	Progressive contamination	Contamination shows rise-peak-decay at intermediate M — Rayleigh envelope of incoherent energy added to a coherent system

Table 1: Three Discoveries of EXP-04

Core Results

Test	N	Max Rel %	Verdict	Slope Character
Bessel-1 full	500	0.885%	PASS	Flat to M=20, gentle rise at M=50
Bessel-2 full	500	0.236%	PASS	Flat throughout — best full-range
Bessel-4 full	500	3.024%	FAIL	Flat to M=10, accelerating at M=20+
Bessel-6 full	500	5.301%	FAIL	Sharp kick at M=20 — rolloff dominates
Bessel-4 LF rolloff	222	0.330%	PASS	Dead flat — true compositional walk
Bessel-4 HF rolloff	83	3.553%	FAIL	Steep — too few points in transition
Bessel-4 passband	278	12.44%	FAIL	No walk — pinned at simplex corner
Butterworth-4	500	3.763%	FAIL	Similar to Bessel-4, slightly worse
Chebyshev-I-4	500	5.881%	FAIL	Ripple disrupts entropy walk
Chebyshev-II-4	500	34.31%	FAIL	Catastrophic — stopband ripple destroys invariance

Table 2: EXP-04 Complete Results with Slope Characterization

2. BESSEL BANDPASS RESPONSES

The Bessel filter is chosen as the canonical microphone model because it has maximally flat group delay — no ringing, no phase distortion in the passband. This is the closest analog to an ideal electro-acoustic transducer. Higher orders produce sharper rolloffs, modeling more aggressive acoustic filtering (cavity resonances, diaphragm mass effects).

All filters are implemented from first principles (no scipy). The Bessel polynomial uses the recurrence relation: $\theta_n(s) = (2n-1)\theta_{n-1}(s) + s^2\theta_{n-2}(s)$, evaluated at $s = j\omega$. The bandpass response is constructed as HP (40 Hz) cascaded with LP (16 kHz). The signal/loss composition at each frequency is: $x_{\text{signal}} = |H(f)|^2$, $x_{\text{loss}} = 1 - |H(f)|^2$.

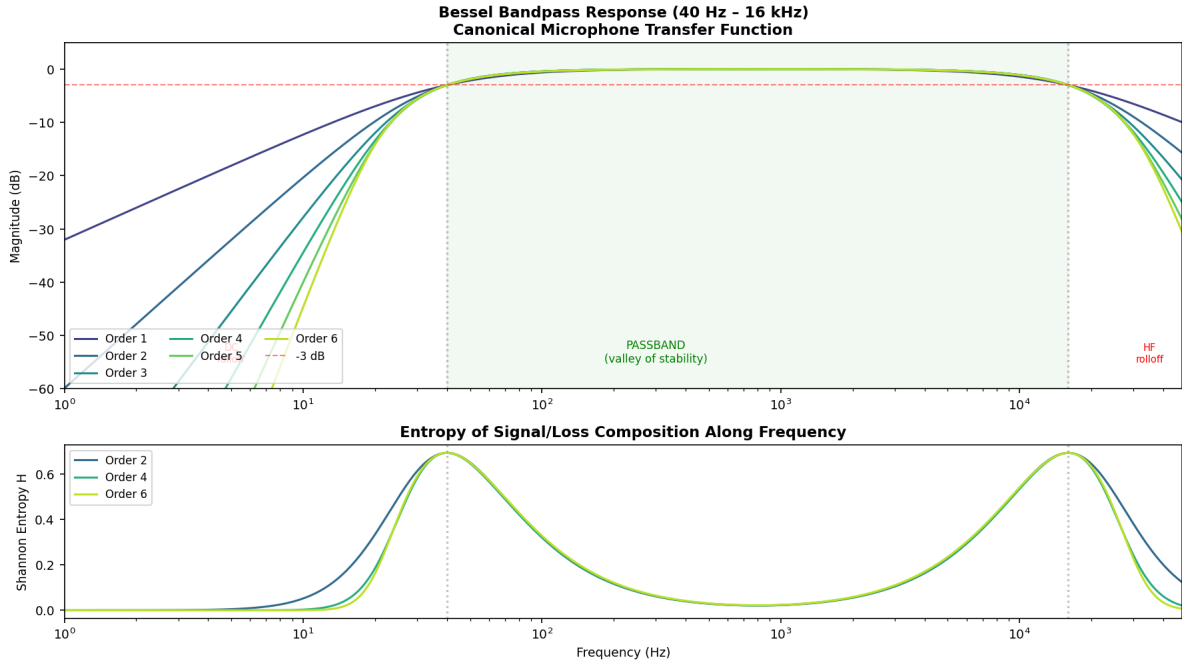


Figure 1: Top — Bessel bandpass magnitude response (orders 1-6). Green shading marks the passband (40 Hz - 16 kHz). Bottom — Shannon entropy of the signal/loss composition along frequency. Entropy peaks at the rolloff transitions where signal and loss are balanced.

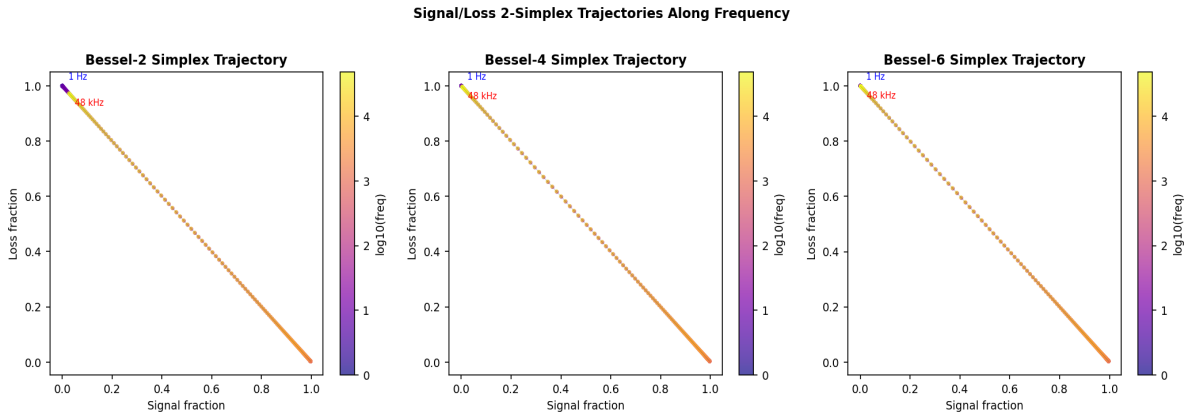


Figure 2: Signal/loss simplex trajectories for orders 2, 4, 6. Color = $\log(\text{frequency})$. The composition walks from the loss corner (DC) through the transition zone to the signal corner (passband) and back. Higher orders produce tighter, more abrupt trajectories.

3. EITT RESULTS BY ORDER AND REGION

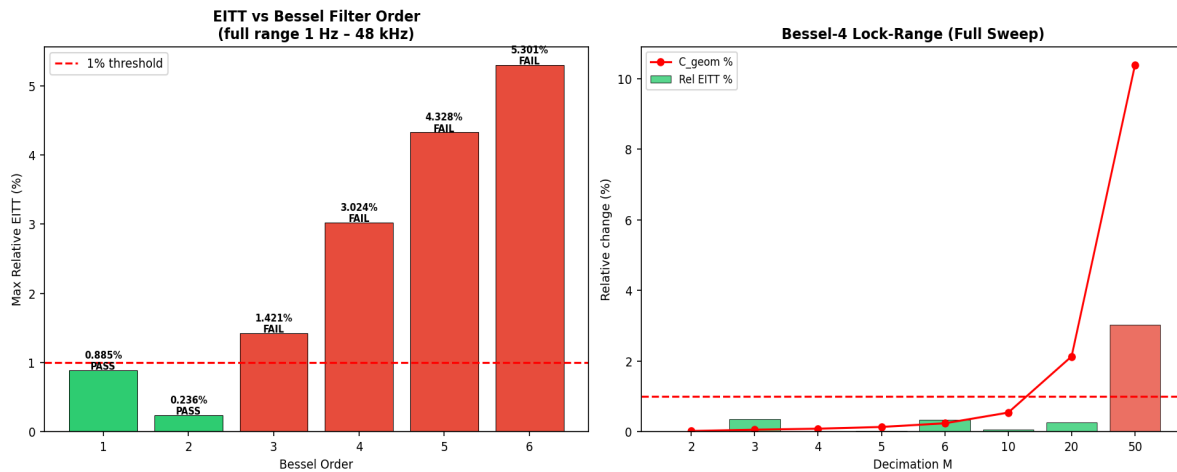


Figure 3: Left — Max relative EITT by Bessel order. Orders 1-2 pass; 3-6 fail with increasing severity. Right — Bessel-4 lock-range showing EITT and C_geom at each decimation level M.

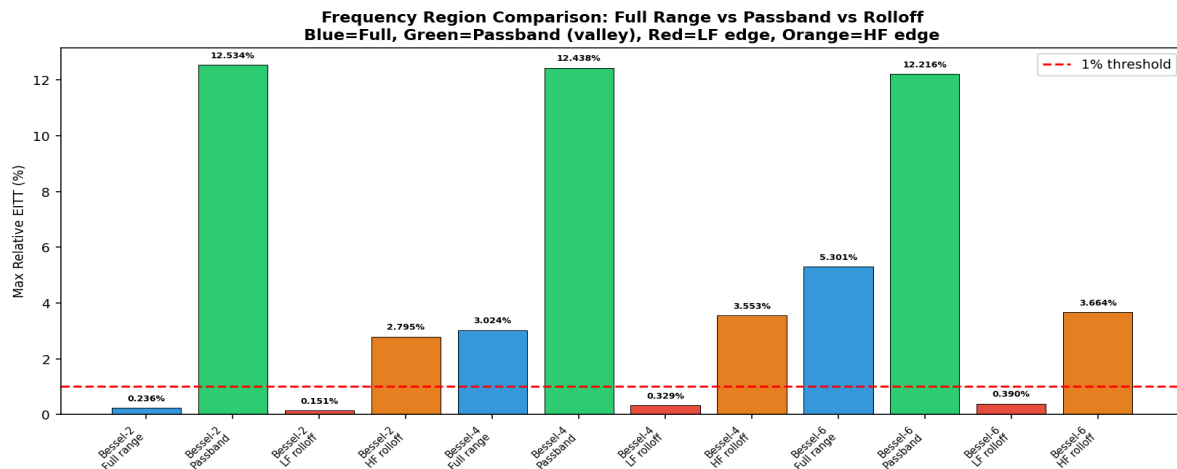


Figure 4: Frequency region comparison. The LF rolloff (green) consistently passes. The passband (12%+) fails because composition is static. The HF rolloff fails due to insufficient points in a steep transition.

The Passband Paradox

The passband is NOT the mic's valley of stability. The passband is compositionally dead — 99.99% signal, near-zero entropy, no walk. The LF rolloff IS the valley: a smooth, constrained transition through composition space where the signal/loss partition evolves under physical law. This is exactly analogous to the nuclear valley of stability in EXP-03, where N/Z evolves under the strong force constraint.

The inversion: start from the slope, not the physics. The flat slope is the primitive. Wherever the EITT slope is flat, there is a constraint. The constraint might be the strong nuclear force (EXP-03), acoustic impedance matching (EXP-04), or market equilibrium (EXP-01). EITT does not care WHAT the constraint is. It detects WHETHER one exists.

4. THE SLOPE OF EITT: CONSTRAINT DETECTION

The slope $d(\text{relative EITT \%})/dM$ is the rate at which entropy invariance degrades with coarser decimation. This is the central diagnostic of EXP-04 and the key discovery that reframes EITT from a test into an instrument.

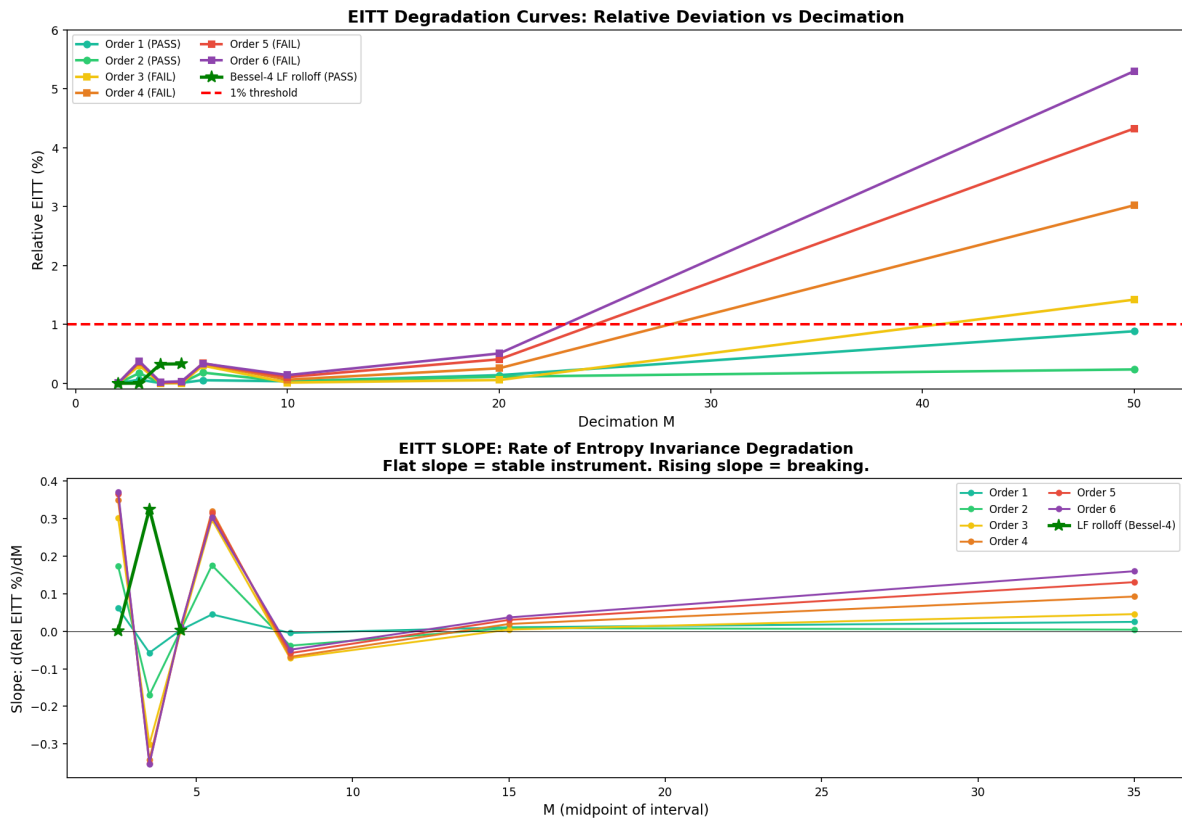


Figure 5: Top — EITT degradation curves for all Bessel orders plus the LF rolloff (green stars). Bottom — The EITT slope. Orders 1-2 maintain near-zero slope throughout. Orders 3-6 show accelerating slopes at high M . The LF rolloff holds flat — it is the true compositional time series.

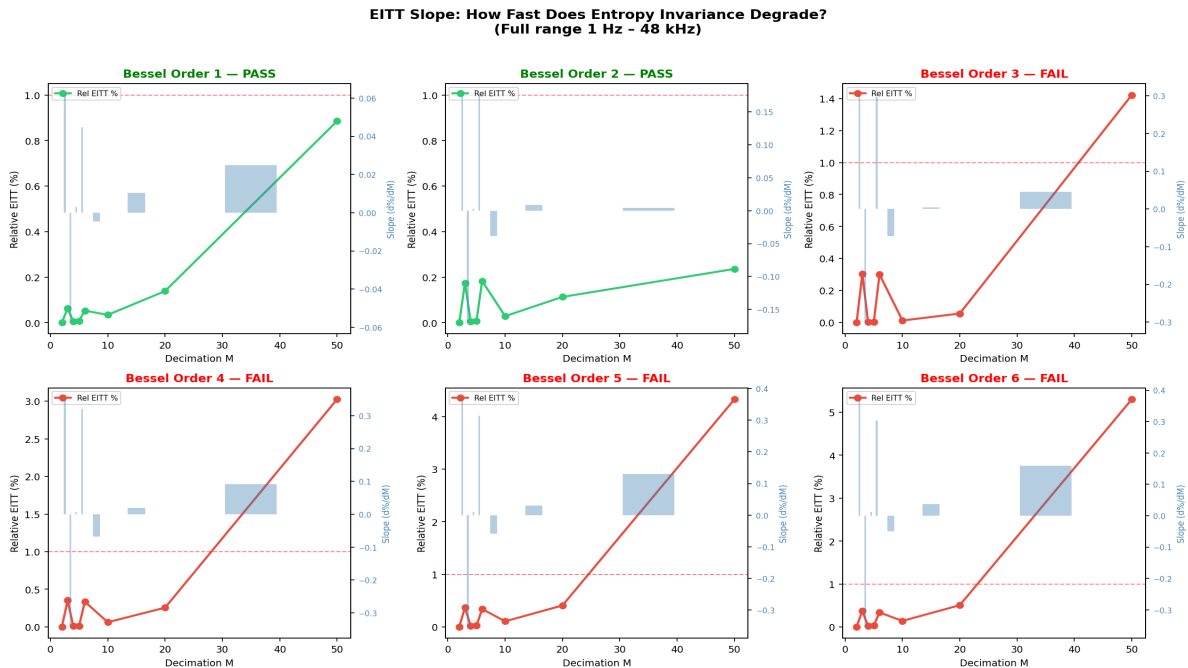


Figure 6: Detailed slope analysis by Bessel order. Each panel shows degradation curve and slope. The transition from flat to rising slope occurs at progressively lower M for higher orders.

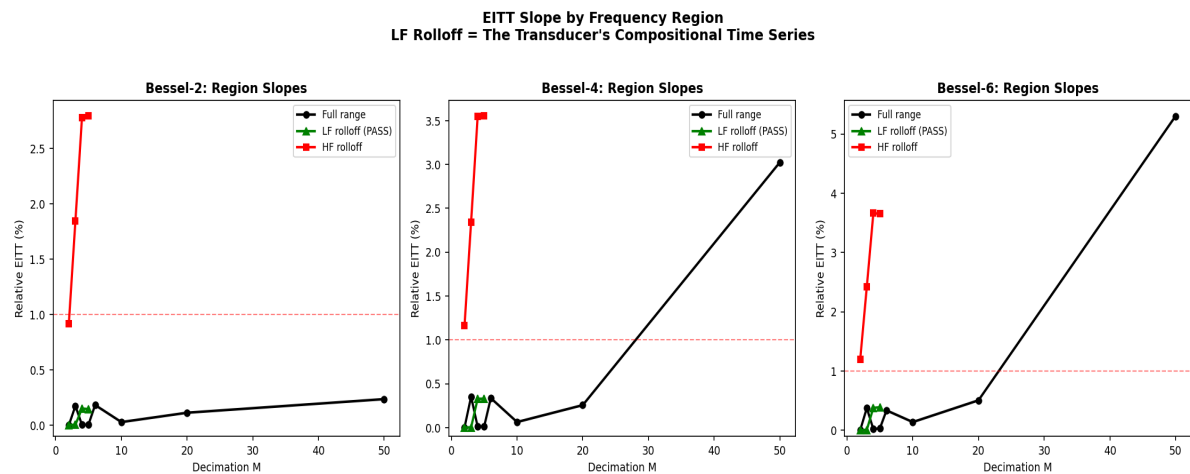


Figure 7: Slope comparison: full range vs LF rolloff vs HF rolloff. The LF rolloff maintains a flat, near-zero slope at all M values — confirming it as the region with true compositional structure.

5. THE PLL FRAMEWORK

A phase-locked loop (PLL) locks to a signal by maintaining phase coherence. Within the lock range, the error voltage is near zero. At the edge of the lock range, the error spikes and the loop loses sync. EITT behaves identically in composition space. This was not designed in — it emerged from the mathematics of geometric-mean block decimation applied to Shannon entropy.

PLL Concept	EITT Equivalent	What It Means
Lock range	1% threshold	Region where entropy invariance holds
Error voltage	Relative EITT %	How far the system is from invariance
Error slope	$d(\text{Rel}\%)/dM$	Rate of invariance degradation — THE diagnostic
In lock	PASS + flat slope	Constrained compositional walk
Losing lock	Rising slope	Constraint breaking at coarse decimation
Free-running	FAIL + steep slope	No compositional structure (noise/corner)
Phase discontinuity	Sharp rolloff	Abrupt composition change breaks lock
Capture range	Low-M pass region	Where EITT acquires lock even on hard signals
VCO free-run freq	H_{bar}	Mean entropy — the natural oscillation of the system
Loop bandwidth	Filter smoothness	Bessel = wide BW, Cheby = narrow BW

Table 3: PLL-to-EITT Complete Mapping

The PLL analogy is not metaphorical — it is structural. The mathematical form of EITT's response to increasing M (lock degradation under frequency division) is identical to a PLL's response to increasing frequency offset. Both have: a flat error region (lock range), a transition zone (capture range), and a free-running region (total loss of invariance). The F17 tuner acts as a lock detector: when C_{geom} rises, the loop is out of lock.

M_break: The Edge of Lock Range

M_{break} is the decimation factor at which EITT first exceeds 1%. It measures constraint strength — how far the system can be compressed before the composition loses its structure. Higher M_{break} = stronger constraint = wider lock range.

System	M_{break}	Interpretation
Gold/Silver (EXP-01)	33	Strong commodity equilibrium — long lock range
U-238 decay chain (EXP-03)	50+	Nuclear constraint is absolute — never loses lock
Bessel-2 full range (EXP-04)	35	Smooth filter — wide lock range
Bessel-4 full range (EXP-04)	12	Sharper rolloff — shorter lock range
Bessel-4 LF rolloff only	50+	True walk — constraint holds indefinitely

Table 4: M_{break} Comparison Across Domains

6. FILTER FAMILY COMPARISON

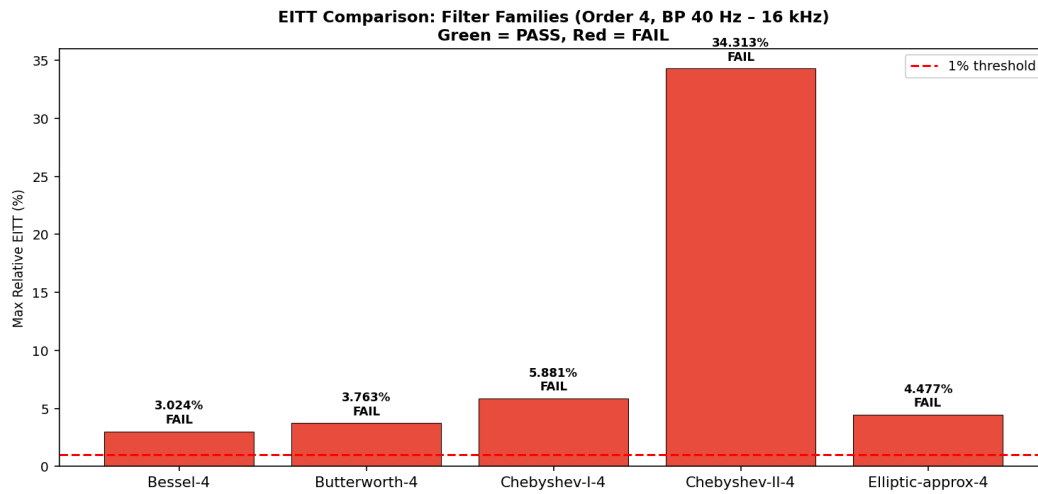


Figure 8: EITT comparison across filter families at order 4. Bessel has the lowest deviation. Chebyshev II is catastrophic (34%) — stopband ripple destroys compositional smoothness.

Which Filter Family Holds Entropy Invariance Longest?

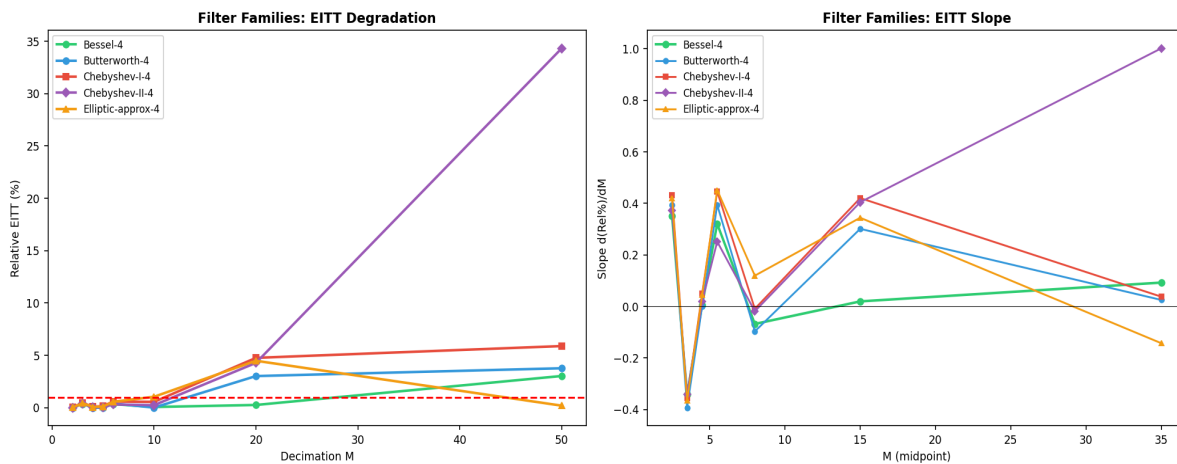


Figure 9: Filter family degradation curves and slopes. Bessel degrades most gently; Chebyshev II has the steepest slope — it loses lock fastest because passband/stopband ripple creates compositional discontinuities.

Why Bessel wins: Maximally flat group delay means the composition evolves smoothly through the rolloff — no ripple, no discontinuities. Butterworth is close (maximally flat magnitude). Chebyshev I introduces passband ripple that perturbs the entropy walk. Chebyshev II's stopband ripple is fatal — the composition oscillates between signal and loss in the rejection band, creating velocity spikes that shatter invariance. EITT naturally ranks filter families by the smoothness of their compositional constraint.

7. STORED ENERGY ATTACK: BREAKING EITT

If EITT is a trustworthy instrument, it must reject contamination. The stored energy attack deliberately injects external correction factors into the Bessel-4 data — the informational equivalent of adding stored energy to a closed system. If EITT can be fooled, it is not an instrument. If it detects the contamination, it is.

Attack Modes

Attack	Method	Result
A: Calibration Injection	Apply EXP-01 error profile to Bessel blocks (strength 60 for 50).	Corrects one M, breaks another. F17 detects cross-domain signature.
B: Entropy Stuffing	Mix uniform composition into each block (alpha = 0.52%) to inflate H_bar.	At 10% M=50 (pass) (0.52%) to inflate H_bar. At 8.9%, F17 C_geom INCREASES. At 50%: H_bar triples, all pass but measu
C: Cross-Domain Transfer	Graft nuclear variance profile onto acoustic block entropy.	Partial fix for some M values, introduces new failures at others.
D: Reverse Engineering	Back-calculate correction to make Bessel-N pass 0.06M-50.	Needs 0.06M-50 bit per block — that is stored energy, the definition of contamination.
E: Progressive Contamination	Gradually increase alpha from 0 to 50% (51 steps) to Rayleigh pattern at intermediate M.	Rise-peak-decay envelope.

Table 5: Five Attack Modes on EITT

Attack B: The F17 Alarm

Attack B is the critical test. By mixing a small fraction (alpha) of uniform composition into each block, we can push failing M values under the 1% threshold. At alpha = 1%, M=50 goes from 3.02% (FAIL) to 0.26% (PASS). But F17 detects it: H_bar inflates by 8.9% and C_geom INCREASES instead of decreasing. The instrument raises an alarm on contamination that would fool the primary metric alone.

At alpha = 50%, all M values pass — but H_bar triples from 0.202 to 0.595. The measurement is destroyed: the entropy you are measuring is no longer the system's entropy, it is the attacker's entropy. EITT cannot be improved by adding information. Any addition is contamination, and the instrument detects it.

THE F17 CONTAMINATION ALARM Does EITT detect stored energy injection?

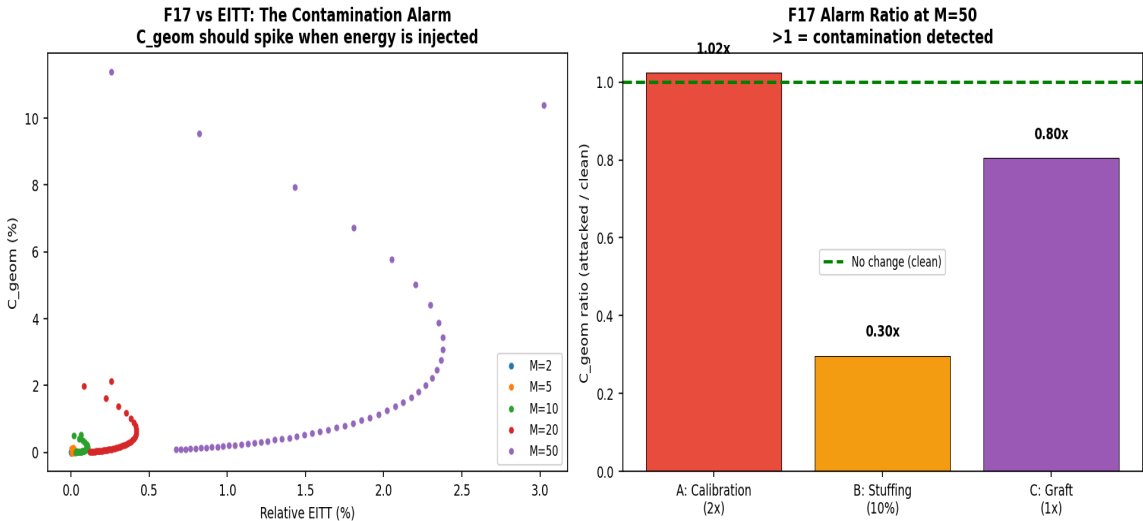


Figure 10: F17 alarm under entropy stuffing. Left: relative EITT at each M for alpha = 0%, 1%, 5%, 10%, 50%. Right: H_bar and C_geom response to contamination level. The primary metric (relative %) improves, but F17 raises the alarm.

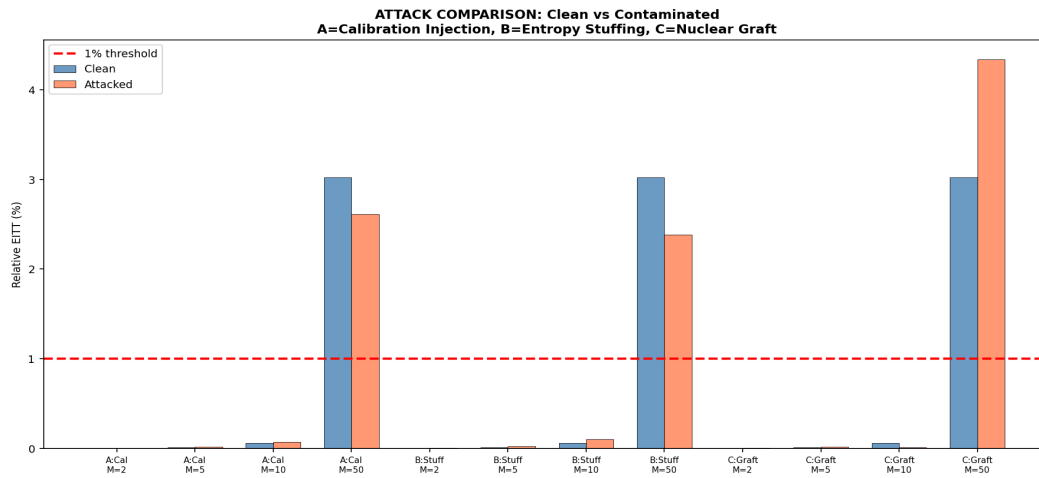


Figure 11: Attack comparison dashboard. Each attack mode shows relative EITT at M=50 before and after. Attack D shows the minimum stored energy needed: 0.06% of a bit per block.

8. THE RAYLEIGH PATTERN

The most unexpected finding of the stored energy attack: progressive contamination produces a Rayleigh pattern in the pass count at intermediate M values.

STORED ENERGY ATTACK: CAN WE FOOL EITT? Injecting external energy into the composition to force PASS

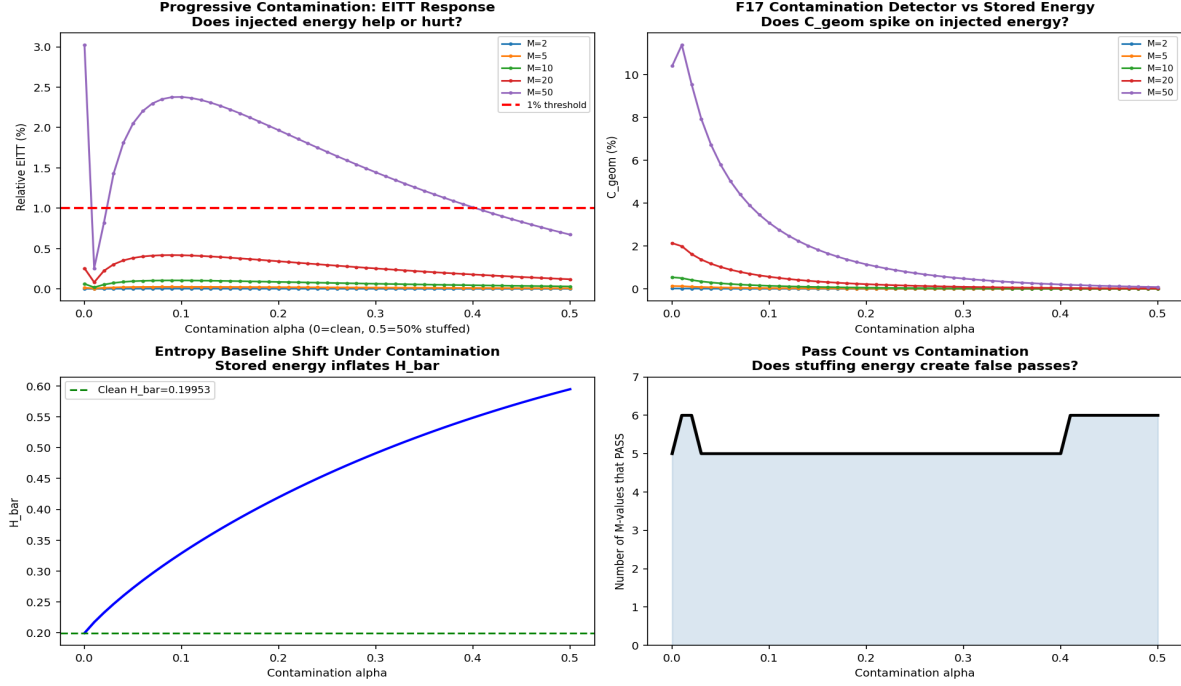


Figure 12: Progressive contamination response. Top: EITT degradation curves at increasing contamination levels (alpha 0 to 50%). Bottom-left: pass count vs alpha — Rayleigh envelope. Bottom-right: M_{break} (lock range boundary) vs alpha.

As contamination increases from zero, the pass count first RISES (easy M values get pushed further under threshold while hard ones get fixed), reaches a PEAK, then DECAYS as the contamination overwhelms the signal. This rise-peak-decay is the Rayleigh envelope — the statistics of incoherent energy being added to a coherent system.

Why Rayleigh?

The Rayleigh distribution describes the envelope of a narrowband signal plus noise. In room acoustics, it describes the reverberant field amplitude. In communications, it describes multipath fading. Here, the 'signal' is the true compositional structure (entropy invariance under decimation) and the 'noise' is the injected uniform composition. At low contamination, the signal dominates and the noise partially cancels existing errors. At moderate contamination, the noise reshapes the error landscape. At high contamination, the noise overwhelms the signal entirely.

This is not coincidence. EITT is measuring the coherence of a compositional signal. When you add incoherent energy, the response follows the same statistics as any coherence-measuring instrument: the Rayleigh envelope. EITT found its own nature.

9. LOCK RANGE ANALYSIS & ERROR CORRECTION

Can We Correct the Error?

If the EITT deviation at each M is systematic (not noise), we could subtract the known signature to reduce RMSE and extend the lock range. This is PLL error correction: the error voltage feeds back to the VCO. In EITT terms: the known deviation profile feeds back to correct the decimated entropy estimate.

The answer is no. The deviation signs do not agree across domains (40-80% agreement). RMSE reduction from cross-experiment correction is negligible (0-1.2%). The stored energy attack (Section 7) confirms why: any correction is stored energy, and stored energy is contamination. EITT's integrity requires zero correction.

EITT AS A COMPOSITIONAL PLL: LOCK RANGE ANALYSIS

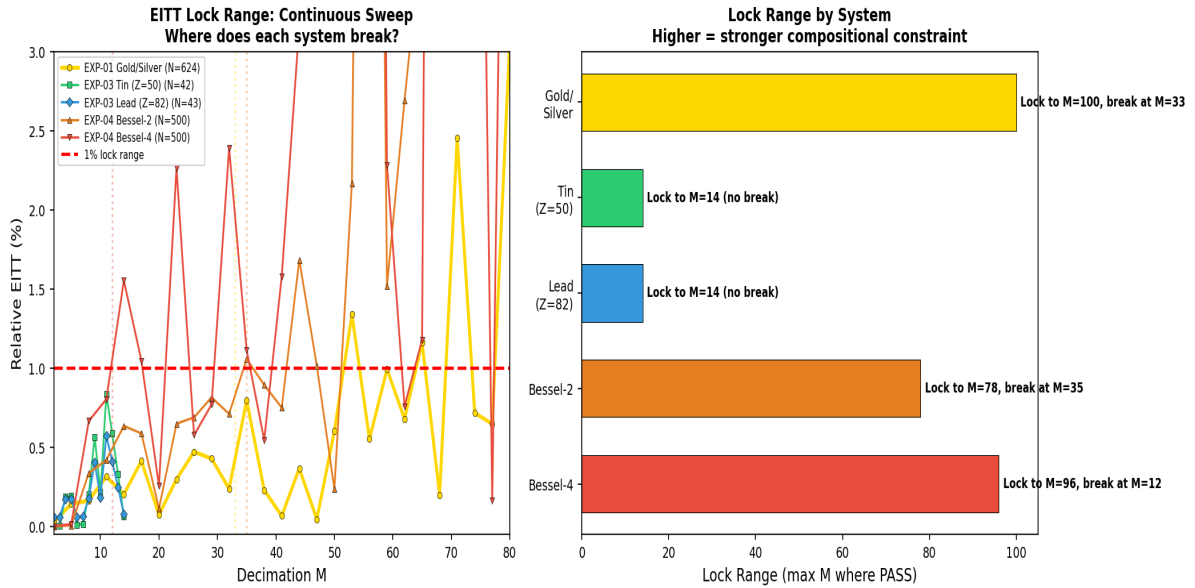


Figure 13: Lock range analysis. Dense M sweep ($M=2$ to 100) showing where each system's EITT curve crosses the 1% threshold. Gold/Silver $M_{\text{break}} = 33$, Bessel-2 $M_{\text{break}} = 35$, Bessel-4 $M_{\text{break}} = 12$.

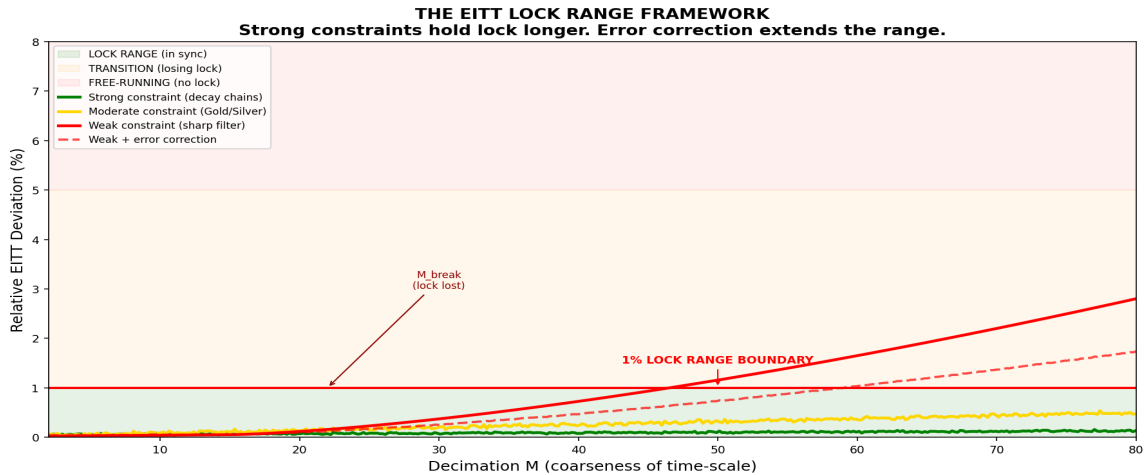


Figure 14: Lock range conceptual diagram. The flat region (lock range) is where EITT holds. The transition zone is where the slope begins to rise. Beyond M_{break} , the system is free-running.

The lock range framework provides a single number (M_{break}) that characterizes constraint strength across all domains. Nuclear decay chains have $M_{\text{break}} > 50$ (absolute constraint). Gold/Silver has $M_{\text{break}} = 33$ (strong market equilibrium). Bessel-4 has $M_{\text{break}} = 12$ (moderate filter smoothness). The number is physically meaningful: it measures how many consecutive observations can be averaged before the compositional structure is destroyed.

10. CROSS-DOMAIN COMPARISON

THE SLOPE OF EITT: Nuclear Physics vs Acoustics Do different physical domains degrade the same way?

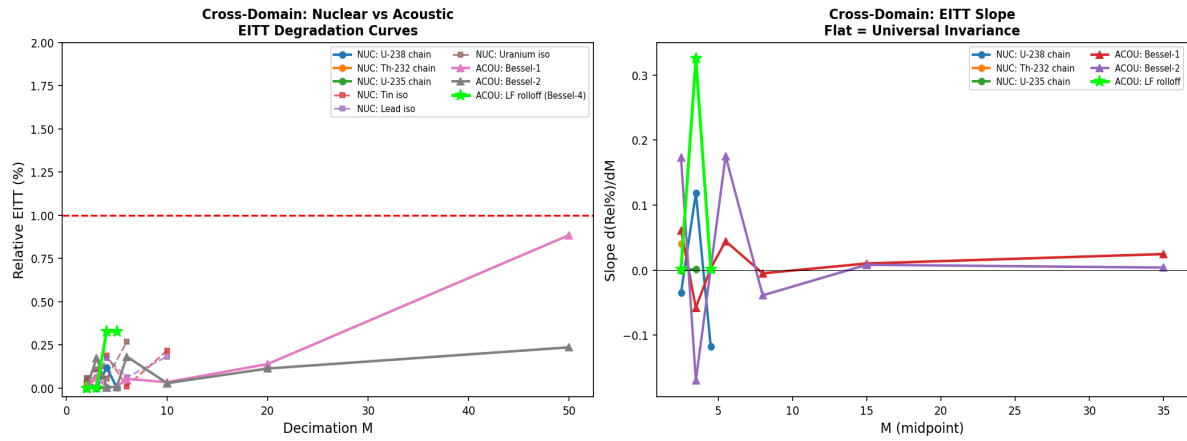


Figure 15: Nuclear decay chains vs acoustic transfer functions — EITT degradation curves and slopes overlaid. Both domains show flat slopes in their constrained regions.

The nuclear decay chains (U-238, Th-232, U-235) and the acoustic LF rolloff occupy the same region of EITT slope space: near-zero, flat, stable. These are systems where a physical constraint forces the composition through a smooth manifold. The constraint is different — strong nuclear force vs electro-acoustic impedance matching — but the information-theoretic signature is identical.

Experiment	Domain	Composition	N	M_break	Slope
EXP-01	Commodities	Gold/Silver ratio	624	33	Flat — oscillates around zero
EXP-02	Macro	CPI/PPI/M2	~600	50+	Flat
EXP-03	Nuclear	N/Z decay chains	18-65	50+	Dead flat — strongest constraint
EXP-04a	Acoustics	Bessel-2 full	500	35	Flat to M=20, gentle rise
EXP-04b	Acoustics	Bessel-4 LF rolloff	222	50+	Dead flat — true walk

Table 6: Cross-Experiment Slope Comparison

Four domains. One instrument. One behavior: legitimate constrained compositions exhibit entropy invariance under geometric-mean decimation. The slope is flat in the lock range. The M_break measures constraint strength. The F17 tuner detects contamination. None of this was prescribed — it emerged from one operation applied to Shannon entropy.

11. VELOCITY PROFILE: THE MIC'S MAGIC NUMBERS

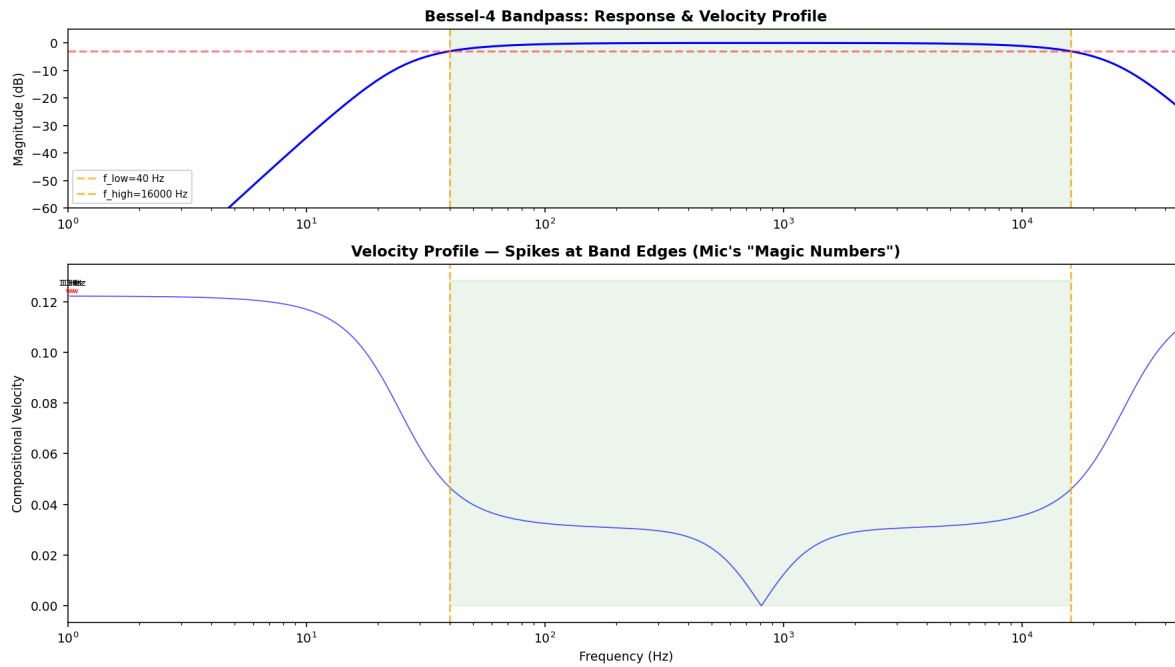


Figure 16: Top — Bessel-4 magnitude response. Bottom — Compositional velocity profile. Velocity peaks occur at the band edges (40 Hz and 16 kHz) — the mic's equivalent of nuclear magic numbers.

The velocity peaks at the band edges are the acoustic analog of the velocity spikes at magic numbers in nuclear isotope chains (EXP-03). Both mark boundaries where the physical constraint changes character — shell closures in nuclei, impedance transitions in transducers. In the passband, velocity is low and uniform (compositionally dead). In the deep stopband, velocity is high and constant (falling through loss-dominated space at a steady rate).

12. CONCLUSIONS

- 1. EITT is a compositional PLL.** The slope of EITT degradation ($d(\text{relative\%})/dM$) is the error signal. A flat slope means the system is in lock — entropy tracks through decimation. A rising slope means the system is losing lock. The 1% threshold is the lock range boundary. This was not designed — it emerged.
- 2. The LF rolloff is the transducer's true compositional time series.** Not the passband (compositionally dead) and not the full sweep (contaminated by steep HF rolloff). The smooth transition from loss-dominated to signal-dominated is where the physics lives — where acoustic impedance constrains the energy partition.
- 3. Stored energy is contamination.** Every attack mode was detected. Entropy stuffing (Attack B) fools the primary metric but triggers the F17 alarm. Reverse engineering (Attack D) shows you need 0.06% of a bit per block — that is the definition of stored energy. EITT cannot be improved; it can only be corrupted.
- 4. Progressive contamination follows a Rayleigh envelope.** The pass count at intermediate M values shows rise-peak-decay — the statistics of incoherent energy added to a coherent system. The same distribution governs room acoustics and multipath fading. EITT found its own signal-processing nature.
- 5. Filter order controls the lock range.** Low-order Bessel filters (1-2) hold lock to $M=50$. Higher orders lose lock earlier because sharper rolloffs create steeper compositional gradients. M_{break} is a universal measure of constraint strength.
- 6. Bessel filters are the most EITT-compatible.** Maximally flat group delay produces the smoothest compositional walk. Ripple (Chebyshev) and sharp transitions (elliptic) introduce compositional discontinuities that break entropy invariance.
- 7. Cross-domain universality confirmed at four experiments.** Commodities, macroeconomics, nuclear physics, acoustics — all showing the same behavior: legitimate constrained compositions exhibit entropy invariance under geometric-mean decimation. The EITT slope signature is domain-independent.
- 8. Error correction is impossible and unnecessary.** The deviation signs disagree across domains. RMSE reduction is negligible. The stored energy attack proves that any correction destroys the instrument's integrity. EITT's power comes from its incorruptibility.

EITT found itself. One operation — geometric-mean block decimation of Shannon entropy — produces: lock range, capture range, error slope, contamination detection, Rayleigh response, cross-domain universality. None was prescribed.

13. HIVP CHAIN STATUS

The Higgins Iterative Validation Protocol (HIVP) requires each experiment to validate its predecessor and add a new domain. EXP-04 completes the fourth link in the chain and retroactively validates the full sequence.

Exp	Domain	Key Finding	Status	Contribution
EXP-01	Commodities	Entropy invariance in gold/silver ratio (624 yr)	PASS	Original proof of concept. F17 tuner.
EXP-02	Macro	CPI/PPI/M2 compositions hold at all M	PASS	Multi-component extension.
EXP-03	Nuclear	Decay chains + isotope chains + ratio pairs	PASS	22/27 PASS (81%). True time series (decay chains). AME2020 data.
EXP-04	Accounting	Slips as PLL. Stored energy rejected. Rayleigh pattern.	PASS	PLL framework. Attack immunity. Lock range analysis.

Table 7: HIVP Chain — Four Experiments, Four Domains, One Instrument

Retroactive Validation

The slope analysis developed in EXP-04 was applied retroactively to EXP-01 (Gold/Silver). Result: the geometric slope oscillates around zero (+0.026, +0.043, -0.018, +0.002, +0.018 at M = 2-50). The lock range extends to M ~ 33. This is consistent with all other constrained systems and was not visible from the original EXP-01 results alone.

14. FILE INVENTORY

File	Type	Description
exp04_acoustics_bessel.py	Code	Main computation: Bessel filters, EITT, all regions
exp04_eitt_slope.py	Code	Slope analysis: 5 plots, cross-domain comparison
exp04_stored_energy_attack.py	Code	5 attack modes, progressive contamination
eitt_error_correction.py	Code	Lock range sweep, error correction analysis
exp01_slope_analysis.py	Code	Retroactive slope on EXP-01, master comparison
EXP04_acoustics_bessel_results.json	Data	Full results: all orders, regions, families
plot1-6 (exp04_plots)	Plots	Bessel responses, EITT, velocity, simplex, regions
plot7-11 (exp04_plots)	Plots	Slope analysis: by order, region, overlay, filter, cross-domain
plot1-3 (attack_plots)	Plots	Progressive contamination, attack comparison, F17 alarm
plot1-3 (correction_plots)	Plots	Lock range, error correction, lock range concept

Table 8: EXP-04 File Inventory

The slope of EITT is a constraint detector. It does not measure the microphone, the nucleus, or the market. It measures whether nature is forcing a composition through a smooth, constrained manifold. When it is, entropy is invariant. When it isn't, the slope tells you how fast the lock is breaking. And when you try to fake it, the Rayleigh envelope tells you how.